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Lab 5

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05.1g: Storage, IAM

2. GCP Cloud Storage #1 (USGS)

- **What role is attached to the Compute Engine default service account?**
  - Editor
- **Would it be sufficient for the VM to perform its functions (i.e. creating buckets and reading/writing objects in them)?**
  - Yes, the editor role provides a lot of permissions allow VM to perform its functions
- **What permissions are given by the default access scope to Cloud Storage?**
  - The default access scope provides read-only access to Cloud Storage.
- **Would they be sufficient for the VM to perform its functions (i.e. creating buckets and reading/writing objects in them)?**
  - No, the VM need read/write access to cloud storage to performs its functions such as creating buckets.

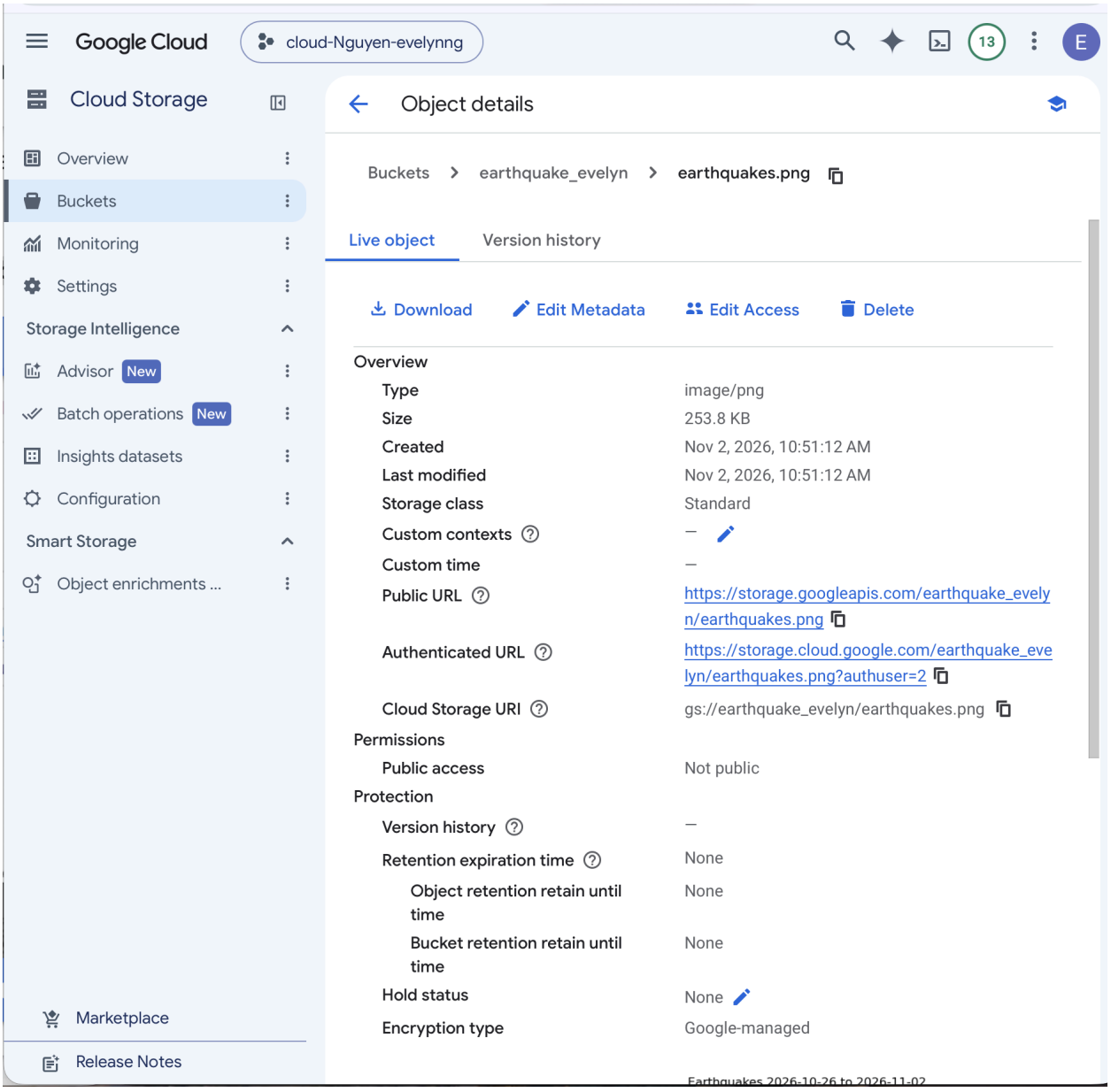
- **What settings are possible for setting the VM's access to the Storage API?**
- 2. Allow full access to all Cloud APIs – Gives the VM full access scopes for all Google Cloud APIs.
- Set access for each API, then find Cloud Storage and select Read Write.

4. USGS data and setup

- **What time did the latest earthquake happen?**
  - 2026-11-02T18:19:19.960Z
- **What was the magnitude (mag)?**
  - 1.7950411434367
- **Where was the place it happened?**
  - "4 km SSW of Brawley, CA"

5. Python plotting code

- **Take a screenshot of the image that has been created for your lab notebook.**



9. Service account roles (Compute)

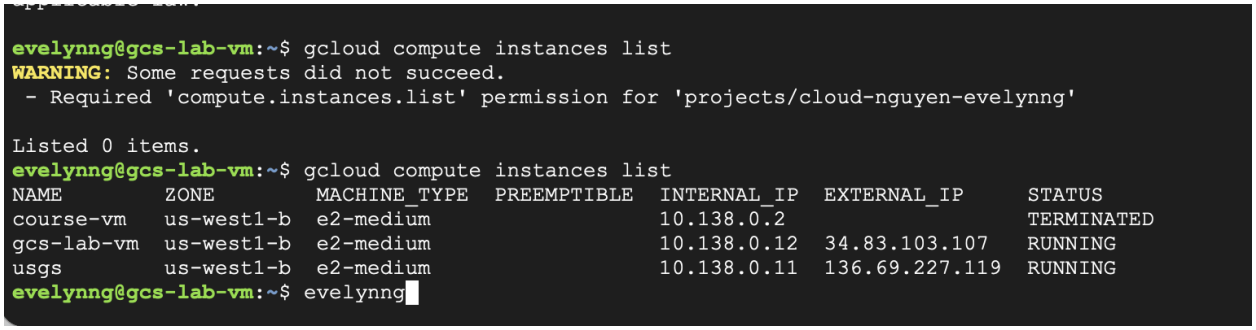
- **What is the exact error message that is returned?**

WARNING: Some requests did not succeed.

- Required 'compute.instances.list' permission for 'projects/cloud-nguyen-evelynng'

Listed 0 items

- **What role needs to be added to the service account's permissions for the VM to have access to list the project's Compute Engine resources?**
    - Compute viewer
- **Take a screenshot of the output for your notebook**



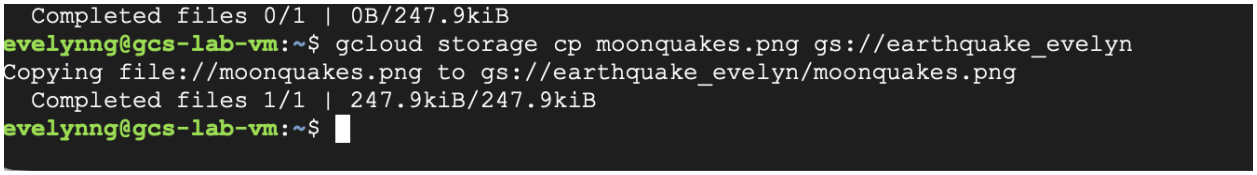
10. Service account roles (Storage)

- **What is the exact error message that is returned?**

⠿ERROR: [gcs-lab@cloud-nguyen-evelynng.iam.gserviceaccount.com] does not have permission to access b instance [earthquake_evelyn] (or it may not exist): gcs-lab@cloud-nguyen-evelynng.iam.gserviceaccount.com does not have storage.objects.create access to the Google Cloud Storage object. Permission 'storage.objects.create' denied on resource '//storage.googleapis.com/projects/_/buckets/earthquake_evelyn/objects/moonquakes.png' (or it may not exist). This command is authenticated as gcs-lab@cloud-nguyen-evelynng.iam.gserviceaccount.com which is the active account specified by the [core/account] property.

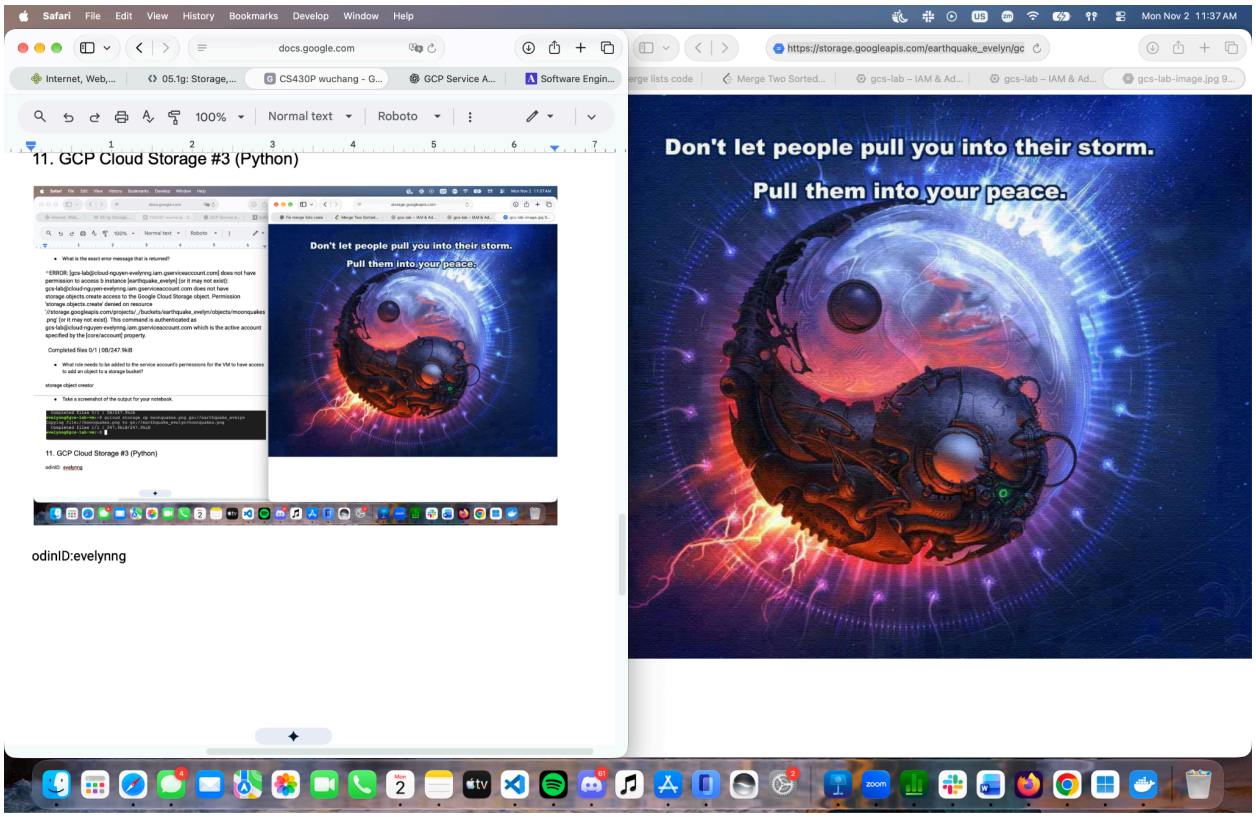
Completed files 0/1 | 0B/247.9kiB

- **What role needs to be added to the service account's permissions for the VM to have access to add an object to a storage bucket?**
- storage object creator
- **Take a screenshot of the output for your notebook.**



13. View object

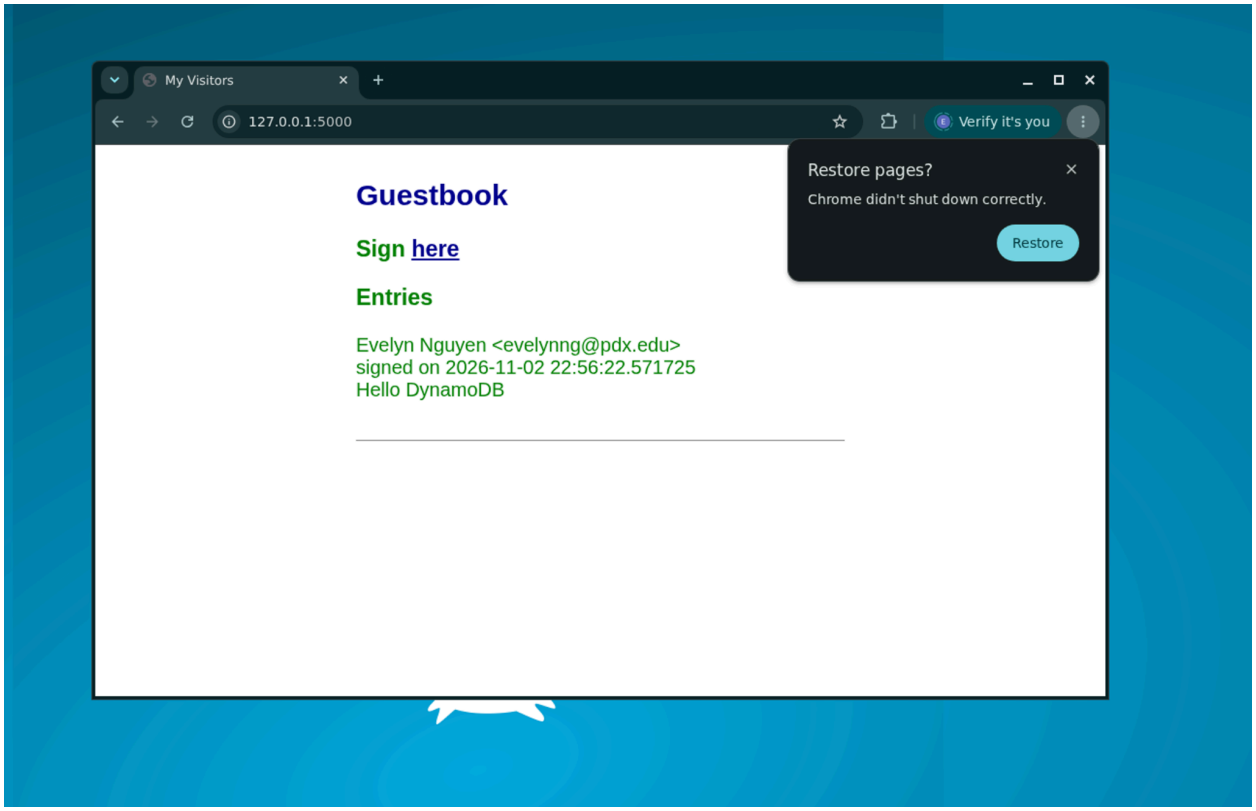
- **Take a screenshot the shows the entire URL and the image that has been retrieved:**



05.2a: DynamoDB guestbook

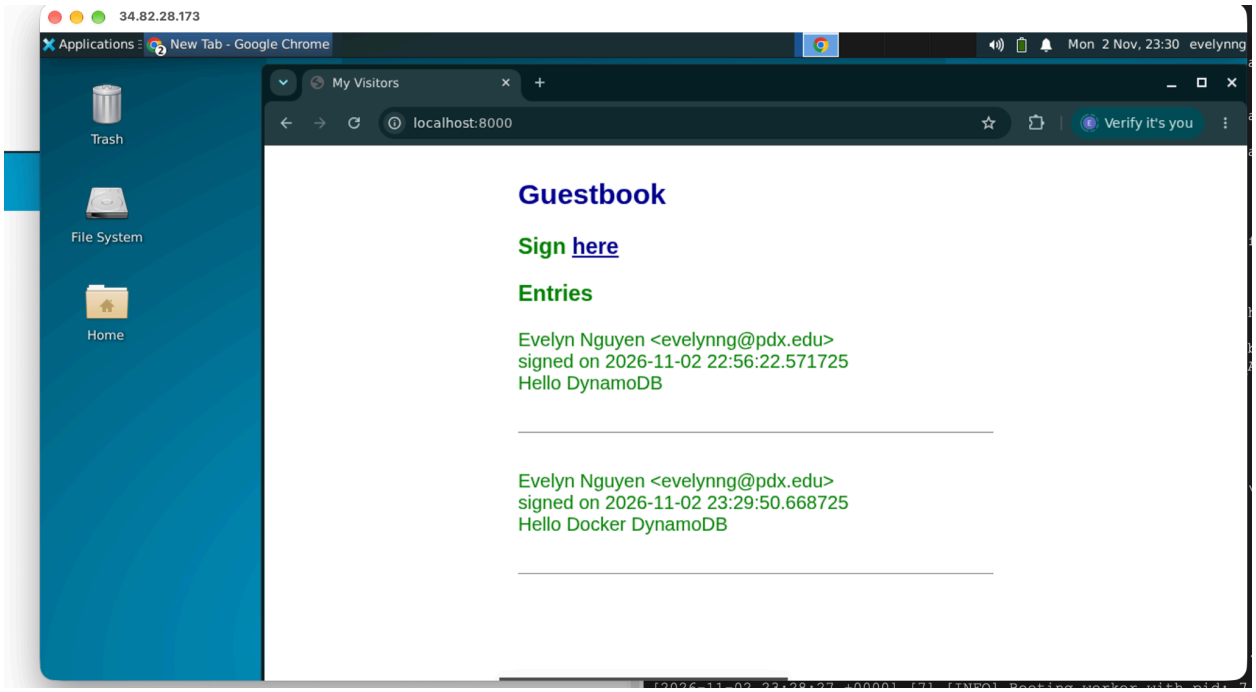
5. Run the application

- **Take a screenshot of the output for your lab notebook.**



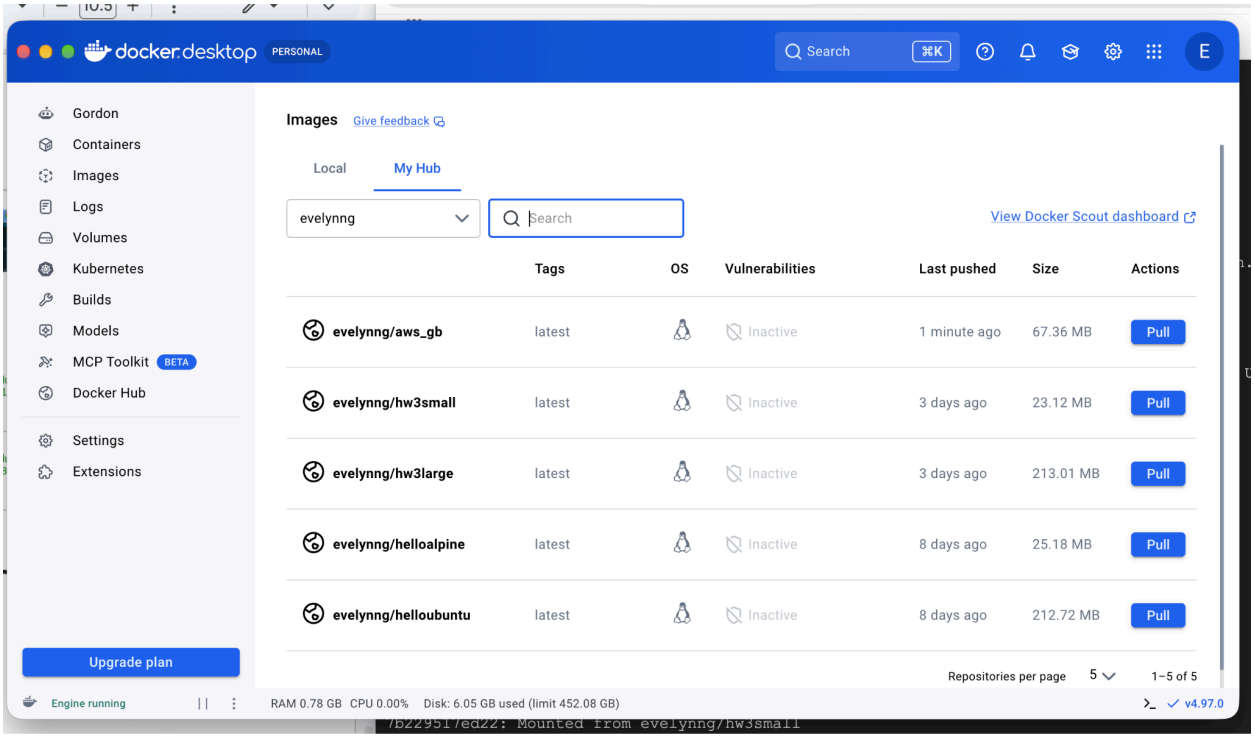
7. Run the application

- **Take a screenshot of the output for your lab notebook.**



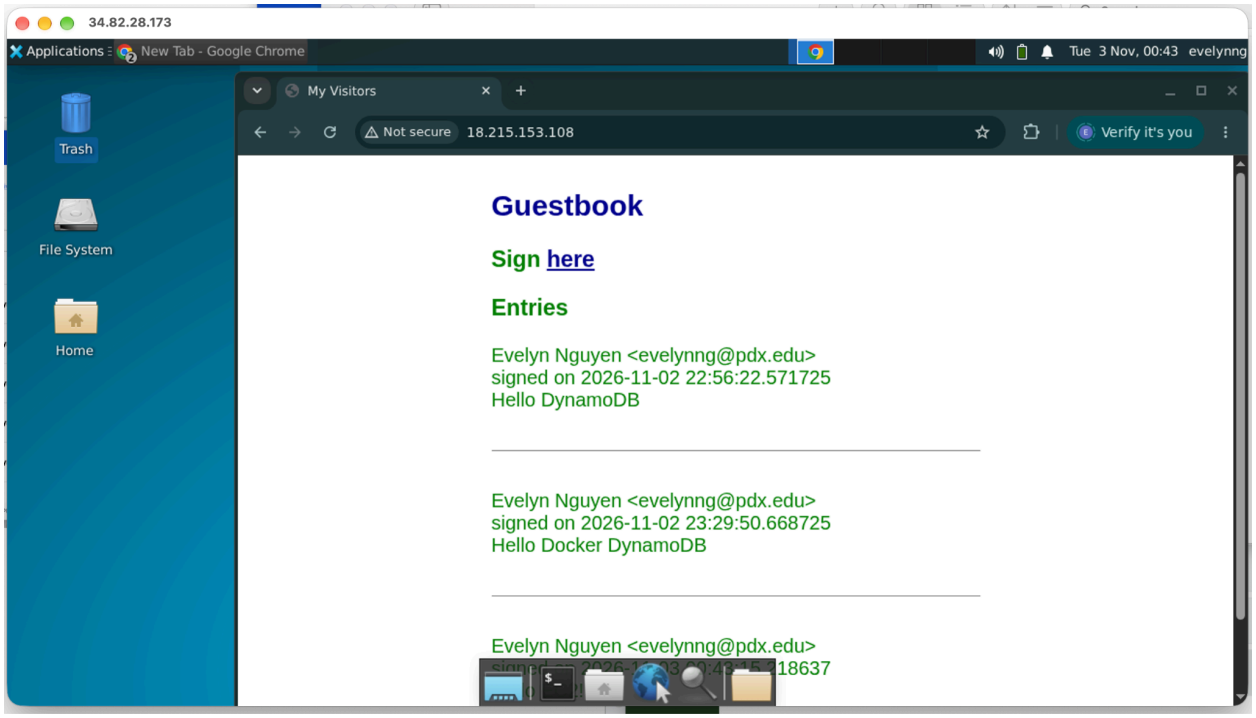
8. Push the container image

- **Take a screenshot of the container image on DockerHub.**



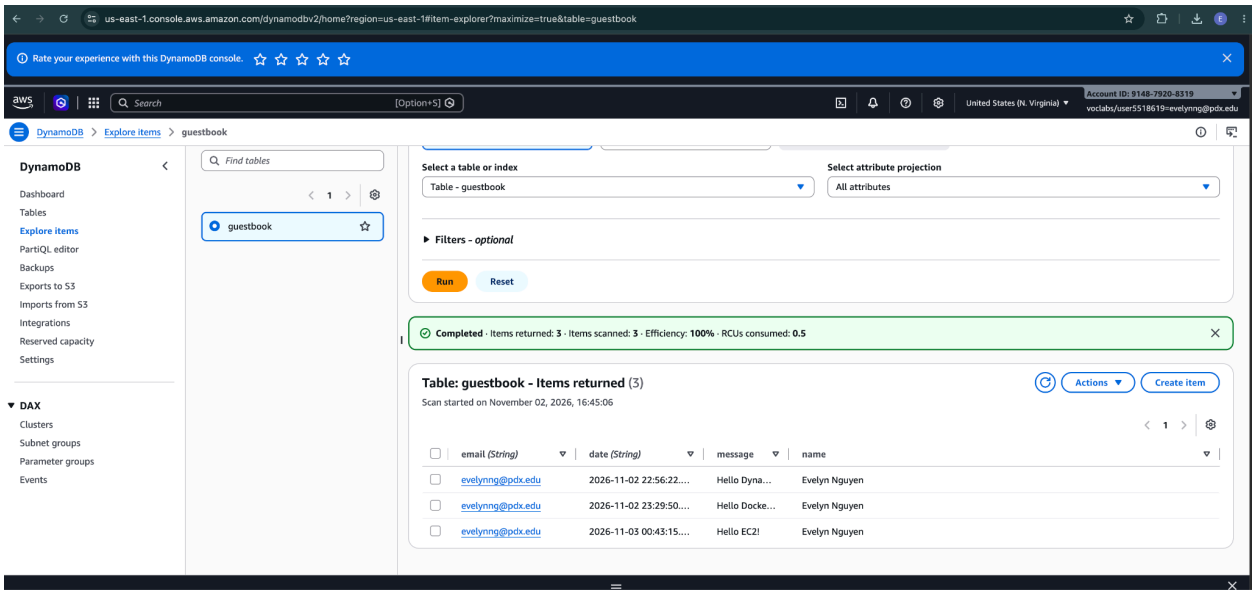
12. Visit the application

- **Take a screenshot as before that shows your entry and the IP address in the URL bar.**



13. View the database

- **Take a screenshot that shows all of the guestbook entries that you added to the DynamoDB table including their timestamps.**



05.2g: Cloud Data

6. Run the application

- **Take a screenshot of the output for your lab notebook.**

